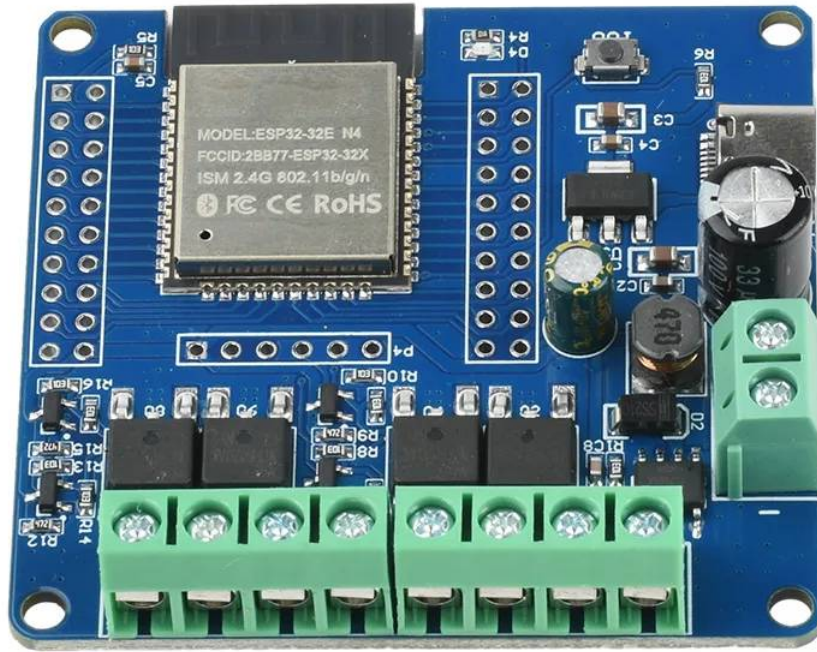


PRODUCT INFORMATION

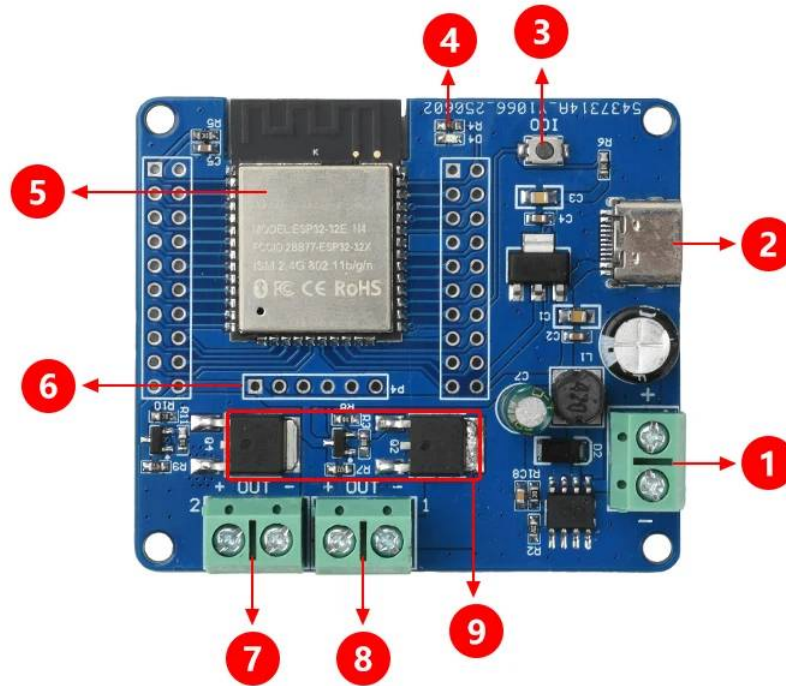
Recommend A Super Practical ESP32 Development Board — A Programmable And Resettable Development Board With Relay Module, Specifically Adapted To Smart Home Control Scenarios. The Core Of This Board Is The ESP32-32E Module, Equipped With Both WiFi And BLE Functions, And All I/O Ports Are Fully Led Out, Making Development Particularly Convenient. In Terms Of Power Supply, It Is Very Flexible, Supporting Type-C USB Port 5V Power Supply, Or You Can Choose The DC-5-60V To 5V Power Supply Circuit. Here, It Is Recommended That You Prioritize Using DC-12V Power Supply For Better Stability. Whether You Are A Beginner Who Wants To Learn ESP32 Secondary Development, Or Want To DIY Smart Home And Do Wireless Control Projects, This Board Can Handle It. Functional Features: 1. Equipped with a mature and stable ESP32-32E module, featuring a large-capacity 4M Byte Flash 2. USB port + DC 5-60V to 5V power conversion circuit (recommended power supply: DC-12V) 3. Dual-channel and quad-channel MOS switch modules, with 2 specifications for your choice 4. Reserve UART pins to support UART programming development 5. Compatible with the ESP-IDF development environment 6. N-channel MOSFET with a drain-source voltage of 60V 7. Product dimensions: Dual path: 50*57.5mm, hole diameter: 3mm Four-way: 55*57.5mm, hole diameter: 3mm Package Include: Switch module X1 10P double row pin X2 6P single row pin X1 Shorting cap X1 Finally, There Is A Small Reminder To Tell Everyone: This Product Is Mainly For Everyone To Do Development Learning And DIY Accessories. After All, It Involves Some Technical Knowledge. It Is Recommended That You Carefully Read The Product Function Introduction First, Confirm That It Meets Your Needs Before Purchasing, So That It Will Be More Convenient To Use.

PRODUCT IMAGES

ESP32 **2/4-way** MOS switch module



Interface Introduction



1.DC 12V input

2.USB power supply

3.Mode switch button

4.Mode status light

5.ESP32 module

**6. Burning port: IO0 and GND shorted
power can enter the burning mode**

7. Second output

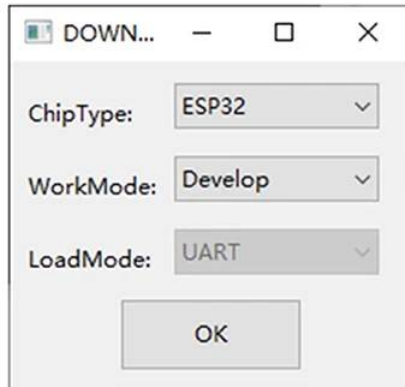
8.First output

9.MOS tube

Firmware Burn-in Process

1.Tool Preparation: USB to Serial Module

2. Steps: Open Flashtool tool, select the chip model ESP32



3. Tick the burn option and specify the firmware path.

4. Short GND-IO0 before power on to enter the burning mode.

5.Connect the device according to the following wiring.

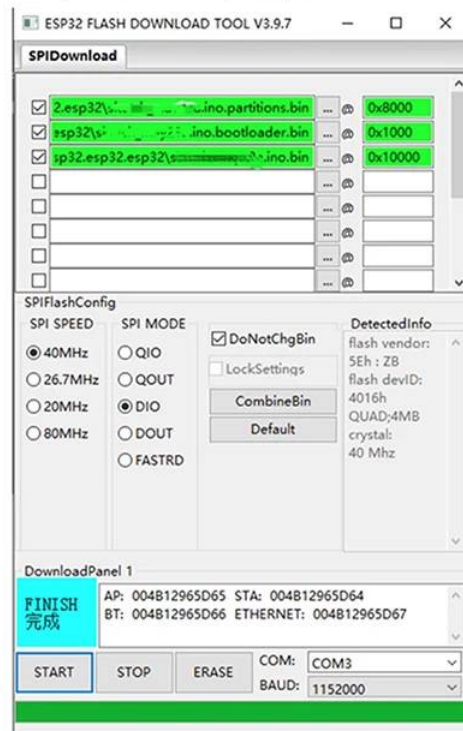
RX-TX

TX-RX

GND-GND

3V3-3V3

6. Click START to start burning, wait for the completion of the prompts



Module Test Methods

1.The module is divided into three modes, through the IO0 button to switch mode:

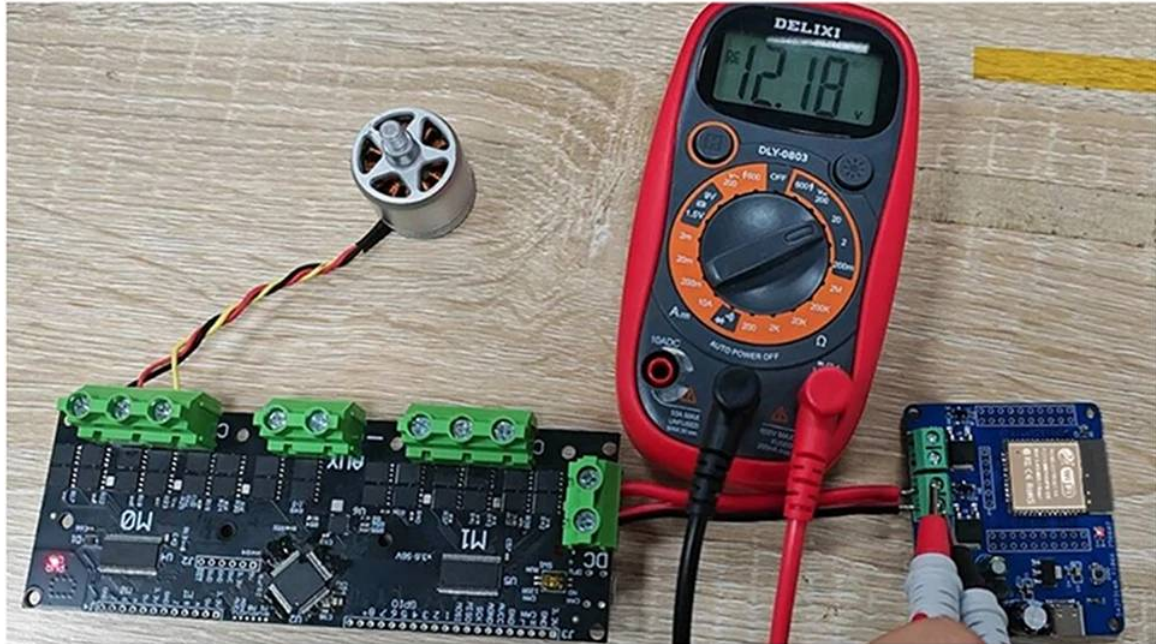
First Mode:Power-on default first mode, the two-way MOS switch are off state LED lights in the off state; at this time, press the button, switch to the second mode.

Second mode: in this mode, the first MOS switch is on, the second MOS is off, the LED light is always on at 50% brightness; at this time, press the button to switch to the third mode.

Third Mode: In this mode, both MOS switches are on, and the LED light is always on at 100% brightness; press the button and switch to the first mode, and so on.

2.Mode test is as follows.

Input 12V DC power supply press the button to switch to the second mode, at this time the first MOS tube open, the output voltage is equal to the input voltage of 12V (pay attention to distinguish between positive and negative output terminal), connected to the load can also work normally.

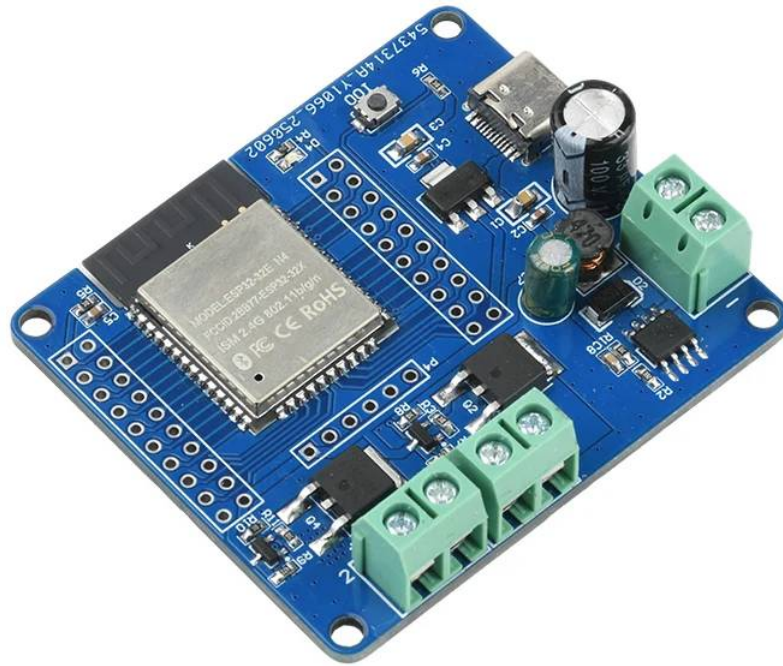


3, Secondary development proposal.

Example program only for functional verification if you need other features please modify the program according to the actual situation of the user can be extended through the interface more control logic recommended in the secondary development framework for the development of complex applications.

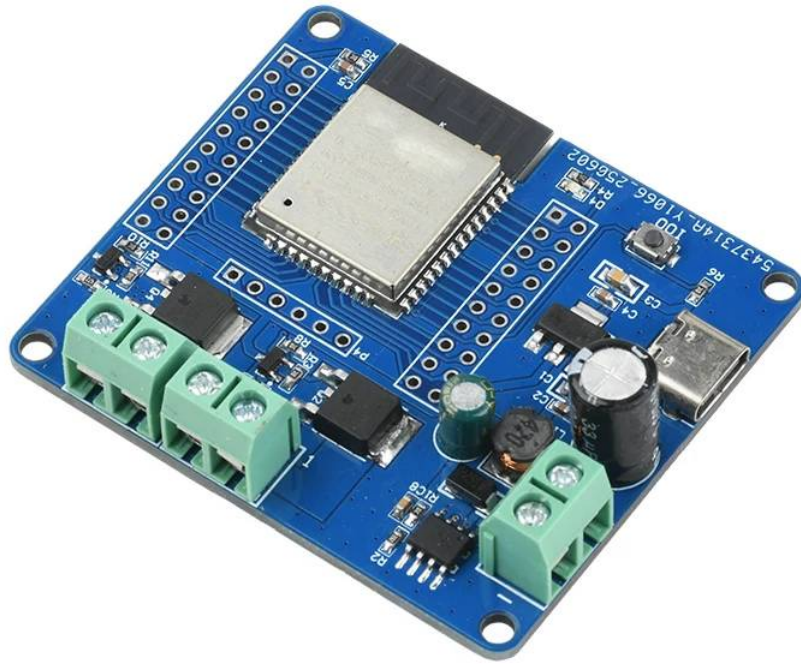
Support for secondary development

Onboard ESP32-32E Module



Suitable for a wide range of power supply environments

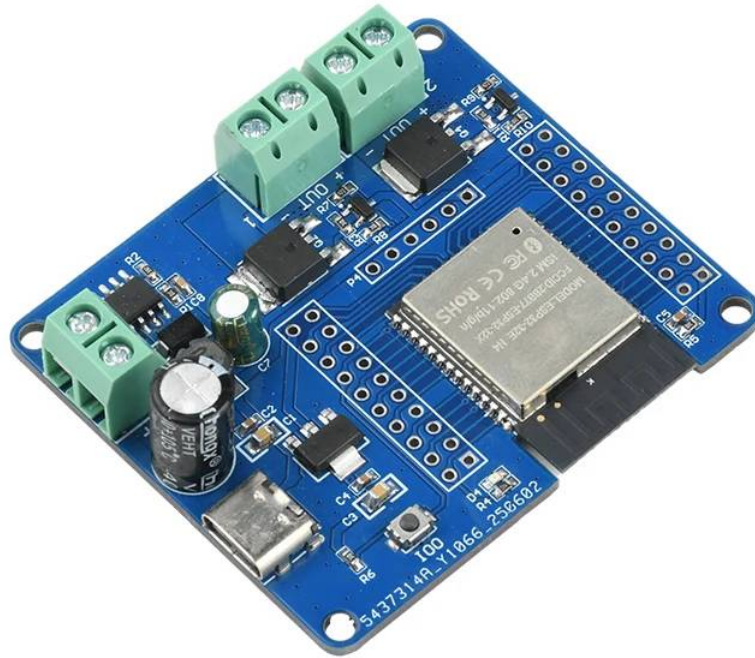
Support DC 5-60V wide voltage input



Flexible options

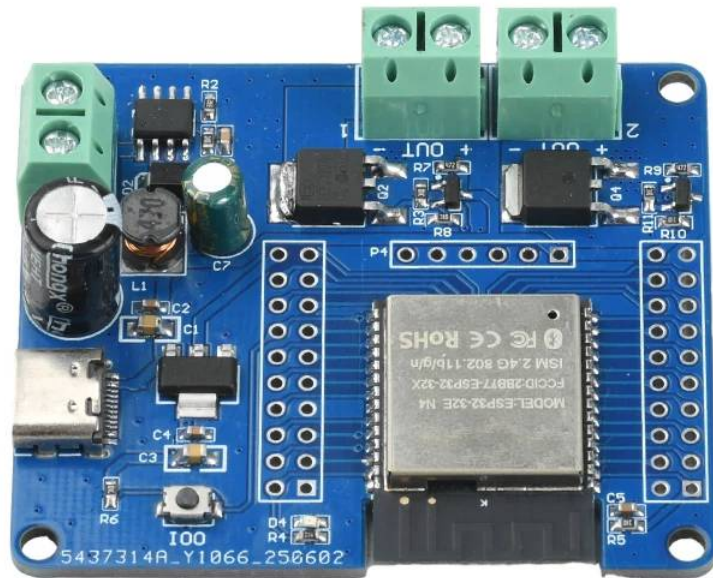
Dual or quad MOS switching modules available





Reserved UART pins

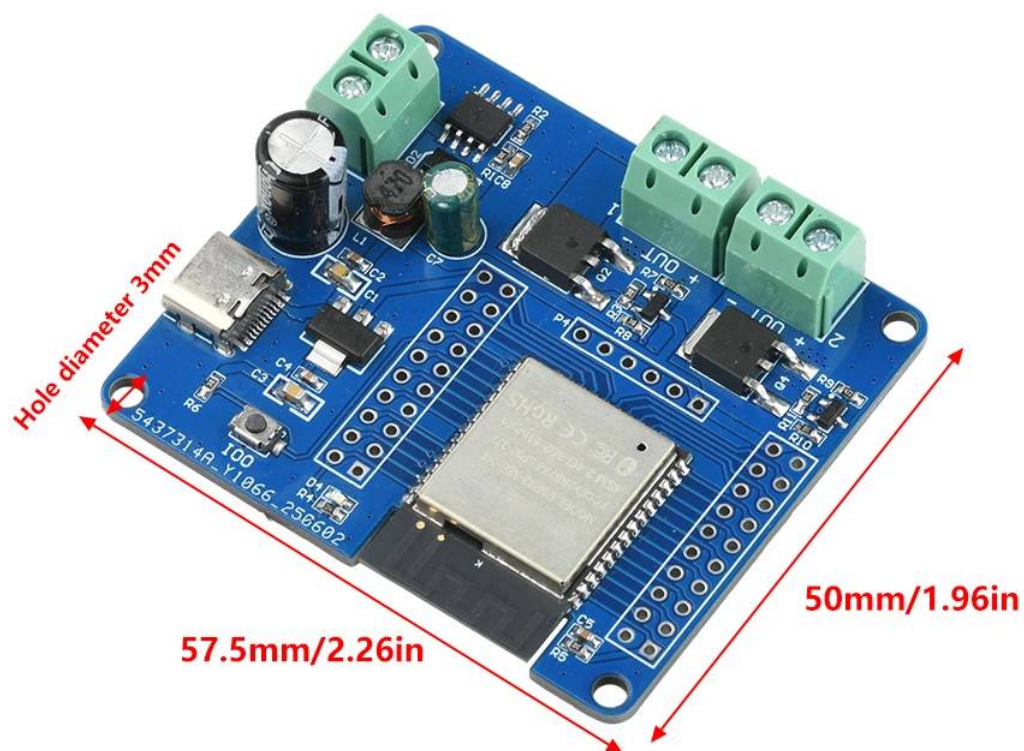
Easy to burn in



Suitable for smart home applications

Compatible with ESP-IDF development environment

Product Size



**DC Powered ESP32 Dual MOS
Switch Module**

Product Size



**DC Powered ESP32 Quad MOS
Switch Module**

